

Disproof of conjecture 00000002192

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Verdict: FALSE

We disprove conjecture 00000002192 as stated. The refutation is unconditional and completely elementary. The standard example S_2 is a height-2 poset on $n = 4$ elements whose order dimension is exactly 2, whereas the conjecture predicts $\lceil \log_2 \log_2 4 \rceil = 1$. The conjecture claims that the bound “follows from the deep nesting of standard examples of random posets”, and it is precisely the standard example that refutes it.

1 The conjecture, quoted verbatim

From `conjectures/00000002192.md`:

English. Definition: The dimension of a poset is the minimal number of linear extensions whose intersection is the poset. Conjecture: The maximal dimension of height-2 posets is $\lceil \log_2 \log_2 n \rceil$ (exact); the bound follows from the deep nesting of standard examples of random posets.

The conjecture is filed in both English and Chinese; the Chinese original is reproduced verbatim (in Unicode) in `README.md`, and the two versions are identical in content. The PDF font does not embed CJK glyphs, so the Chinese is kept in the markdown copy rather than duplicated here.

Throughout, a *linear extension* of a poset P is a strict total order extending the strict order of P , and the *intersection* of linear extensions $L_1, \dots, L_d$ is the relation $x < y \iff (x <_{L_i} y \text{ for all } i)$. The *dimension* $\dim(P)$ is the least d for which some d linear extensions have intersection exactly P .

2 n is never defined

The statement is ill-posed before it is false: the symbol n is never introduced. We show that every natural reading fails.

Reading of n	Counterexample	Formula vs. truth
number of elements $ P $	S_2 has $n = 4$ elements	$\lceil \log_2 \log_2 4 \rceil = 1$ vs. $\dim = 2$
number of minimal elements	S_k has k minimal elements	$\lceil \log_2 \log_2 k \rceil < k$ vs. $\dim = k$
width (largest antichain)	S_k has width k	$\lceil \log_2 \log_2 k \rceil < k$ vs. $\dim = k$
rank / height	every S_k has height 2	$\lceil \log_2 \log_2 2 \rceil = 0$ vs. $\dim = k$

So no choice of n saves the conjecture. The primary witness is the first row, the natural reading $n = |P|$.

3 The standard example S_2

Definition 1 (Standard example). For $k \geq 1$ the *standard example* S_k is the height-2 poset with k minimal elements $a_0, \dots, a_{k-1}$ and k maximal elements $b_0, \dots, b_{k-1}$ and relations

$$a_j < b_i \iff i \neq j.$$

For $k = 2$ take the four elements 0, 1 (minimal) and 2, 3 (maximal), with the two relations

$$0 < 3 \quad \text{and} \quad 1 < 2.$$

Lemma 2. S_2 has height 2, i.e. it contains a 2-element chain but no 3-element chain.

Proof. The chain $0 < 3$ exists. The only relations are the disjoint pairs $0 < 3$ and $1 < 2$. If $x < y < z$ were a 3-chain, then y would be above x and below z ; but the top element of each relation (3 or 2) is maximal and the bottom element (0 or 1) is minimal, and the two relations share no element, so no such y exists. $\square$

4 Two linear extensions whose intersection is S_2

Define the two total orders

$$L_1 = [0, 3, 1, 2] \quad (\text{order } 0 < 3 < 1 < 2), \quad L_2 = [1, 2, 0, 3] \quad (\text{order } 1 < 2 < 0 < 3).$$

Lemma 3. L_1 and L_2 are linear extensions of S_2 and their intersection is exactly the relation of S_2 . Hence $\dim(S_2) \leq 2$.

Proof. Both orders place 0 before 3 and 1 before 2, so both respect the two relations. Conversely, in the intersection:

- the pair $\{0, 1\}$ has $0 < 1$ in L_1 and $1 < 0$ in L_2 , so neither order survives;
- the pair $\{2, 3\}$ has $3 < 2$ in L_1 and $2 < 3$ in L_2 , so neither order survives;
- $0 < 3$ and $1 < 2$ hold in both orders, so both survive.

Hence the intersection is exactly $\{(0, 3), (1, 2)\}$, the relation of S_2 . $\square$

5 No single linear extension realises S_2

Lemma 4. *No single linear extension of S_2 has intersection equal to the relation of S_2 . Hence $\dim(S_2) \geq 2$.*

Proof. Let L be a single linear extension and suppose its intersection (which is just its own total order) equals the relation of S_2 . A total order must compare the incomparable pair $\{0, 1\}$: either $0 <_L 1$ or $1 <_L 0$.

- If $0 <_L 1$, then $0 < 1$ lies in the intersection, but 0 and 1 are incomparable in S_2 (both are minimal, and $0 < 1$ is not one of the two relations) — contradiction.
- If $1 <_L 0$, then $1 < 0$ lies in the intersection, but again 0 and 1 are incomparable — contradiction.

So no single extension realises S_2 . □

Theorem 5. $\dim(S_2) = 2$.

Proof. Combine $\dim(S_2) \leq 2$ (Lemma 3) with $\dim(S_2) \geq 2$ (Lemma 4). □

6 The contradiction

S_2 has $n = 4$ elements, height 2 (Lemma 2), and dimension 2 (Theorem 5). The conjecture asserts that the maximal dimension of a height-2 poset on n elements is

$$\lceil \log_2 \log_2 n \rceil = \lceil \log_2 \log_2 4 \rceil = \lceil \log_2 2 \rceil = \lceil 1 \rceil = 1.$$

But S_2 is a height-2 poset on 4 elements with dimension $2 \neq 1$. Therefore the conjectured value is strictly too small and the conjecture is false.

Corollary 6. *There is a height-2 poset on $n = 4$ elements whose dimension exceeds $\lceil \log_2 \log_2 4 \rceil$. Hence the maximal dimension of height-2 posets is not given by $\lceil \log_2 \log_2 n \rceil$.*

Remark 7. The failure is not a rounding or boundary effect. The same witness family refutes the claim for every $k \geq 2$; see Section 7.

7 The family S_k and the true growth

Theorem 8 (Dushnik–Miller). *For every $k \geq 1$, $\dim(S_k) = k$.*

Proof of the lower bound. In a realizer, for each i the incomparable pair $\{a_i, b_i\}$ must be reversed by some extension L with $b_i <_L a_i$; otherwise $a_i < b_i$ would belong to the intersection.

One extension cannot reverse two distinct pairs. Suppose L satisfies $b_i <_L a_i$ and $b_j <_L a_j$ with $i \neq j$. Since $i \neq j$, the relations $a_i < b_j$ and $a_j < b_i$ hold in S_k , and every extension respects them. Combining,

$$a_i < b_j <_L a_j < b_i <_L a_i,$$

a cycle in the total order L , impossible.

Hence each of the k pairs $\{a_i, b_i\}$ requires its own extension, so k extensions are necessary. Conversely, the k extensions

$$L_i = [a_j \ (j \neq i)][b_i][a_i][b_j \ (j \neq i)]$$

are linear extensions (every relation $a_j < b_l$ with $j \neq l$ is respected: a_j precedes b_i in the leading block or a_i precedes b_l in the trailing block), and they realise S_k . Therefore $\dim(S_k) = k$. $\square$

Corollary 9. *For every $k \geq 2$, S_k is a height-2 poset on $2k$ elements with $\dim(S_k) = k > \lceil \log_2 \log_2(2k) \rceil$. Thus the conjecture fails for every $k \geq 2$.*

Proof. By Theorem 8, $\dim(S_k) = k$. For $k = 2, 3, 4$ one checks $\lceil \log_2 \log_2(2k) \rceil = 1, 2, 2 < k$; for $k \geq 5$ the doubly logarithmic quantity grows far more slowly than k (e.g. $\lceil \log_2 \log_2(2k) \rceil < k$ already for all $k \geq 5$ by monotonicity and the value at $k = 5$). Hence $k > \lceil \log_2 \log_2(2k) \rceil$ for all $k \geq 2$. $\square$

Theorem 10 (Hiraguchi; tight for standard examples). *For every poset P with $|P| \geq 2$, $\dim(P) \leq \lfloor |P|/2 \rfloor$. Moreover, for $n \geq 4$ the maximum dimension of a height-2 poset on n elements equals $\lfloor n/2 \rfloor$, attained by $S_{\lfloor n/2 \rfloor}$.*

Sketch. Hiraguchi's inequality gives $\dim(P) \leq \lfloor |P|/2 \rfloor$. The standard example $S_{\lfloor n/2 \rfloor}$ has $2\lfloor n/2 \rfloor \leq n$ elements and dimension $\lfloor n/2 \rfloor$ (Theorem 8); adding $n - 2\lfloor n/2 \rfloor \in \{0, 1\}$ isolated elements does not decrease the dimension nor increase the height beyond 2, so the bound is attained whenever $n \geq 4$. $\square$

The true growth is therefore linear, not doubly logarithmic.

7.1 Even smaller witnesses

Poset	elements	height	dimension	formula
2-element chain	2	2	1	$\lceil \log_2 \log_2 2 \rceil = 0$
V (two minima below one maximum)	3	2	2	$\lceil \log_2 \log_2 3 \rceil = 1$
S_2	4	2	2	$\lceil \log_2 \log_2 4 \rceil = 1$

The 2-element chain even breaks the formula before $n = 4$: the formula returns 0, impossible for a non-empty poset, since every non-empty poset has dimension at least 1.

8 Exhaustive and asymptotic tables

8.1 Exhaustive verification for $n \leq 6$

`reproduce.py` enumerates all labelled posets of height ≤ 2 on n elements for $n = 1, \dots, 6$, deduplicates them, and computes the exact order dimension of each by a bitmask search over tuples of linear extensions:

n	number of height- ≤ 2 posets	true max dim	formula $\lceil \log_2 \log_2 n \rceil$
1	1	1	undefined
2	3	2	0
3	13	2	1
4	87	2	1
5	841	2	2
6	11643	3	2

The true maximum is 1, 2, 2, 2, 2, 3 while the formula is 0, 1, 1, 2, 2; the two agree only at $n = 5$, by coincidence.

8.2 Asymptotic comparison

n	true max $\lfloor n/2 \rfloor$	formula $\lceil \log_2 \log_2 n \rceil$
8	4	2
16	8	2
1024	512	4
10^6	500000	5

The true maximum is linear in n ; the conjectured value is the iterated logarithm. Their ratio diverges.

9 Formalisation

The $n = 4$ contradiction is formalised in the Lean 4 project under `lean4/` (core Lean only, `import Std`, no `Mathlib`, no `sorry`): S_2 is modelled on `Fin 4` by `ltS2 x y := (x = 0 ∧ y = 3) ∨ (x = 1 ∧ y = 2)`, a linear extension is a list respecting it, and the intersection of a family is the pointwise intersection of the induced orders. The main statements are

```

theorem dim_le_two : DimLe 2
theorem no_single (L : List E) (h : IsLinExt L) :
  ¬ (∀ x y, Intersection [L] x y ↔ ltS2 x y)
theorem dim_eq_two : DimEq 2
theorem height_two : HeightTwo
theorem formula_at_four : Nat.log2 (Nat.log2 4) = 1
theorem conjecture_00000002192_false :
  HeightTwo ∧ DimEq 2 ∧ Nat.log2 (Nat.log2 4) = 1 ∧ 2 ≠ 1

```

Here `DimLe d` says a realizer of d linear extensions exists and `DimEq 2` says the dimension is exactly 2. Core Lean has no real logarithm, so $\lceil \log_2 \log_2 4 \rceil = 1$ is formalised as the exact integer iterated logarithm `Nat.log2 (Nat.log2 4) = 1`; at $n = 4$ the two readings coincide. See `lean4/README.md` for the statement table, the proof strategy, and the axiom audit.

10 Reproducibility

The exhaustive checks are carried out by `reproduce.py` using the Python 3 standard library only:

```
$ python3 reproduce.py
... (exact dimension of  $S_2, S_3, S_4$ ; the  $S_2$  realizer; the chain and  $V$  witnesses;
exhaustive height- $\leq 2$  enumeration for  $n \leq 6$ ; the asymptotic table)
PASS: all checks verified; conjecture 00000002192 is FALSE.
```

It prints `PASS` and exits 0 exactly when every check holds, and exits non-zero otherwise. The exact order dimension is computed by iterative deepening over tuples of linear extensions encoded as bitmasks of the two orientations of every incomparable pair, with branch and bound; for a given length the search is exhaustive unless pruned by the valid counting bound, so the reported dimensions are exact.

Status against the submission rules

Rule 3 requires a LaTeX source, a PDF document, and a Lean 4 project.

- **LaTeX source** — `main.tex`, a standalone article using `geometry`, `amsmath`, `amssymb`, `amsthm`, `array`, `parskip`, `booktabs`; compiles with `tectonic --outdir build main.tex`.
- **PDF** — at `build/main.pdf`, produced by `tectonic --outdir build main.tex`.
- **Lean 4 project** — `lean4/` with `lean-toolchain` pinned to `leanprover/lean4:v4.33.1`, `lakefile.toml` (package `tlmc2192`, library `Main`, no dependencies), `Main.lean`, `Check.lean` (`#print axioms`), and `lean4/README.md`. `lake build` exits 0 and the audit reports no `sorryAx`.